

Time & Work

Answer Key & Detailed Solutions — Questions 1 to 21

Question 1

Rahul, Rakshita and Gurmeet, working together, would have taken more than 7 days to finish a job. On the other hand, Rahul and Gurmeet, working together would have taken less than 15 days to finish the job. However, they all worked together for 6 days, followed by Rakshita, who worked alone for 3 more days to finish the job. If Rakshita had worked alone on the job then the number of days she would have taken to finish the job, cannot be

A) 20

B) 17

C) 16

✓ D) 21 ← Correct Answer

Solution:

Let the rates of work of Rahul, Rakshita, and Gurmeet be r , s , and g units/day. Let total work be W .

$$1. W/(r+s+g) > 7 \rightarrow r+s+g < W/7$$

$$2. W/(r+g) < 15 \rightarrow r+g > W/15$$

$$3. 6(r+s+g) + 3s = W \rightarrow 6(r+g) + 9s = W$$

$$\text{From (3), substitute } W \text{ into (1): } 6(r+g) + 9s > 7(r+s+g) \rightarrow 2s > r+g.$$

$$\text{From (3), substitute } W \text{ into (2): } 6(r+g) + 9s < 15(r+g) \rightarrow s < r+g.$$

$$\text{Combining: } s < r+g < 2s.$$

$$T_s = W/s = [6(r+g) + 9s]/s = 6(r+g)/s + 9.$$

$$\text{Since } 1 < (r+g)/s < 2: \text{ Lower bound: } T_s > 6(1) + 9 = 15. \text{ Upper bound: } T_s < 6(2) + 9 = 21.$$

$$\text{So } 15 < T_s < 21. \text{ Values 16, 17, 20 are possible. 21 is NOT possible (strictly less than 21).}$$

Question 2

Gautam and Suhani, working together, can finish a job in 20 days. If Gautam does only 60% of his usual work on a day, Suhani must do 150% of her usual work on that day to exactly make up for it. Then, the number of days required by the faster worker to complete the job working alone is

TITA (Type in the answer)

Solution:

$$\text{Let Gautam's rate} = G \text{ and Suhani's rate} = S. \text{ Total work } W = (G+S) \times 20.$$

$$\text{If Gautam does } 0.6G, \text{ Suhani does } 1.5S \text{ to make up: } 0.6G + 1.5S = G + S.$$

$$0.5S = 0.4G \rightarrow G/S = 5/4. \text{ Gautam is the faster worker.}$$

$$\text{Let } G = 5k, S = 4k. \text{ Total work } W = (5k + 4k) \times 20 = 180k.$$

$$\text{Time for Gautam alone} = W/G = 180k/5k = 36 \text{ days.}$$

Question 3

A water tank has inlets of two types A and B. All inlets of type A when open, bring in water at the same rate. All inlets of type B, when open, bring in water at the same rate. The empty tank is

completely filled in 30 minutes if 10 inlets of type A and 45 inlets of type B are open, and in 1 hour if 8 inlets of type A and 18 inlets of type B are open. In how many minutes will the empty tank get completely filled if 7 inlets of type A and 27 inlets of type B are open?

TITA (Type in the answer)

Solution:

Let rate of one A inlet = a units/min, one B inlet = b units/min. Volume = V .

Scenario 1: $(10a + 45b) \times 30 = V \rightarrow V = 300a + 1350b \dots(1)$

Scenario 2: $(8a + 18b) \times 60 = V \rightarrow V = 480a + 1080b \dots(2)$

Equate: $300a + 1350b = 480a + 1080b \rightarrow 270b = 180a \rightarrow b = (2/3)a$.

Substitute into (1): $V = 300a + 1350(2/3)a = 300a + 900a = 1200a$.

Scenario 3: New rate = $7a + 27b = 7a + 27(2/3)a = 7a + 18a = 25a$.

Time = $V / \text{Rate} = 1200a / 25a = 48$ minutes.

Question 4

A tank is emptied everyday at a fixed time point. Immediately thereafter, either pump A or pump B or both start working until the tank is full. On Monday, A alone completed filling the tank at 8 pm. On Tuesday, B alone completed filling the tank at 6 pm. On Wednesday, A alone worked till 5 pm, and then B worked alone from 5 pm to 7 pm, to fill the tank. At what time was the tank filled on Thursday if both pumps were used simultaneously all along?

A) 4:48 PM

B) 4:12 PM

✓ C) 4:24 PM ← Correct Answer

D) 4:36 PM

Solution:

Let emptying time = T pm. Rates: pump A = a , pump B = b units/hour. Volume = V .

Monday: $V = a(8-T) \dots(1)$

Tuesday: $V = b(6-T) \dots(2)$

Wednesday: $V = a(5-T) + b(2) \dots(3)$

From (1) and (3): $a(8-T) = a(5-T) + 2b \rightarrow 3a = 2b \rightarrow a/b = 2/3$.

Let $a = 2k$, $b = 3k$.

From (1) and (2): $2k(8-T) = 3k(6-T) \rightarrow 16-2T = 18-3T \rightarrow T = 2$.

Tank is emptied at 2 pm. $V = 2k(8-2) = 12k$.

Thursday (both pumps): Combined rate = $2k + 3k = 5k$.

Time = $V / \text{rate} = 12k / 5k = 2.4$ hours = 2 hours 24 minutes.

Finish time = 2:00 pm + 2h 24m = 4:24 PM.

Question 5

Ramesh and Ganesh can together complete a work in 16 days. After seven days of working together, Ramesh got sick and his efficiency fell by 30%. As a result, they completed the work in 17 days instead of 16 days. If Ganesh had worked alone after Ramesh got sick, in how many days would he have completed the remaining work?

- A) 14.5
- B) 11
- ✓ C) 13.5 ← Correct Answer
- D) 12

Solution:

Let rates be R (Ramesh) and G (Ganesh). Total work $W = (R+G) \times 16 = 16R + 16G$.

After 7 days together: Work done = $7(R+G)$. Remaining = $9(R+G)$.

After Ramesh gets sick, his rate becomes $0.7R$. They work 10 more days ($17-7=10$).

Remaining work = $(0.7R + G) \times 10$.

Equate: $9(R+G) = (0.7R + G) \times 10$

$9R + 9G = 7R + 10G \rightarrow 2R = G$.

Remaining work = $9(R+G) = 9(R + 2R) = 27R$.

Ganesh's rate $G = 2R$.

Time for Ganesh = $27R / 2R = 13.5$ days.

Question 6

Anu, Vinu and Manu can complete a work alone in 15 days, 12 days and 20 days, respectively. Vinu works everyday. Anu works only on alternate days starting from the first day while Manu works only on alternate days starting from the second day. Then, the number of days needed to complete the work is

- 5) 5
- 8) 8
- 6) 6
- ✓ 7) 7 ← Correct Answer

Solution:

Total work = $\text{LCM}(15, 12, 20) = 60$ units.

Anu's rate = $60/15 = 4$ units/day. Vinu's rate = $60/12 = 5$ units/day. Manu's rate = $60/20 = 3$ units/day.

Schedule: Vinu works every day. Anu on days 1, 3, 5, ... Manu on days 2, 4, 6, ...

2-day cycle:

Day 1 (Anu + Vinu): $4 + 5 = 9$ units.

Day 2 (Manu + Vinu): $3 + 5 = 8$ units.

Total per cycle = 17 units.

After 3 cycles (6 days): $3 \times 17 = 51$ units done. Remaining = $60 - 51 = 9$ units.

Day 7 (Anu + Vinu, rate = 9 units/day): completes remaining 9 units exactly.

Total = 7 days.

Question 7

Amar, Akbar and Anthony are working on a project. Working together Amar and Akbar can complete the project in 1 year, Akbar and Anthony can complete in 16 months, Anthony and Amar can complete in 2 years. If the person who is neither the fastest nor the slowest works alone, the time in months he will take to complete the project is

TITA (Type in the answer)

Solution:

Convert to months: 1 year = 12, 2 years = 24. Total work = LCM(12, 16, 24) = 48 units.

$$a + k = 48/12 = 4 \text{ units/month ... (i)}$$

$$k + n = 48/16 = 3 \text{ units/month ... (ii)}$$

$$n + a = 48/24 = 2 \text{ units/month ... (iii)}$$

Adding all three: $2(a+k+n) = 9 \rightarrow a+k+n = 4.5$.

$$a = 4.5 - (k+n) = 4.5 - 3 = 1.5 \text{ units/month.}$$

$$k = 4.5 - (n+a) = 4.5 - 2 = 2.5 \text{ units/month.}$$

$$n = 4.5 - (a+k) = 4.5 - 4 = 0.5 \text{ units/month.}$$

Fastest = Akbar ($k=2.5$), Slowest = Anthony ($n=0.5$). Middle = Amar ($a=1.5$).

$$\text{Time for Amar} = 48/1.5 = 32 \text{ months.}$$

Question 8

Amal and Vimal together can complete a task in 150 days, while Vimal and Sunil together can complete the same task in 100 days. Amal starts the task and works for 75 days, then Vimal takes over and works for 135 days. Finally, Sunil takes over and completes the remaining task in 45 days. If they all started together, with Amal working every day, Vimal working every second day, and Sunil working every third day, how many days would be required to complete the task?

TITA (Type in the answer)

Solution:

Total work = LCM(150, 100) = 300 units. $a+v = 2$ units/day, $v+s = 3$ units/day.

From actual work: $75a + 135v + 45s = 300$.

Substitute $a = 2-v$ and $s = 3-v$:

$$75(2-v) + 135v + 45(3-v) = 300$$

$$150 - 75v + 135v + 135 - 45v = 300$$

$$15v + 285 = 300 \rightarrow v = 1.$$

So $a = 1$, $v = 1$, $s = 2$ units/day.

New schedule (LCM(1,2,3) = 6-day cycle):

Amal works all 6 days: $6 \times 1 = 6$ units.

Vimal works days 2,4,6: $3 \times 1 = 3$ units.

Sunil works days 3,6: $2 \times 2 = 4$ units.

Total per 6-day cycle = 13 units.

Full cycles: $\text{floor}(300/13) = 23$ cycles. Work done = $23 \times 13 = 299$ units. Time = $23 \times 6 = 138$ days.

Remaining = 1 unit. Day 139: Amal works (1 unit/day) $\rightarrow$ done.

Total = 139 days.

Question 9

Bob can finish a job in 40 days, if he works alone. Alex is twice as fast as Bob and thrice as fast as Cole in the same job. Suppose Alex and Bob work together on the first day, Bob and Cole work together on the second day, Cole and Alex work together on the third day, and then, they continue the work by repeating this three-day roster. Then, the total number of days Alex would have worked when the job gets finished, is

TITA (Type in the answer)

Solution:

Bob's rate = 3 units/day. Alex = 6 units/day. Cole = 2 units/day.

Total work = $3 \times 40 = 120$ units.

3-day cycle: Day 1 (A+B): $6+3=9$. Day 2 (B+C): $3+2=5$. Day 3 (C+A): $2+6=8$. Total = 22 units/cycle.

Full cycles: $\text{floor}(120/22) = 5$ cycles. Work done = 110 units. Time = 15 days.

Remaining = 10 units.

Day 16 (A+B): they do 9 units. Remaining = 1 unit.

Day 17 (B+C): they finish 1 unit. Job done.

Alex worked on Days 1, 3 of each of 5 cycles = 10 days. Plus Day 16.

Total Alex days = $10 + 1 = 11$ days.

Question 10

A group of N people worked on a project. They finished 35% of the project by working 7 hours a day for 10 days. Thereafter, 10 people left the group and the remaining people finished the rest of the project in 14 days by working 10 hours a day. Then the value of N is

150) 150

23) 23

36) 36

140) 140

Solution:

Work is proportional to people $\times$ days $\times$ hours.

Phase 1: 0.35W done by N people, 10 days, 7 hrs/day. Effort = $N \times 10 \times 7 = 70N$.

Phase 2: 0.65W done by $(N-10)$ people, 14 days, 10 hrs/day. Effort = $(N-10) \times 14 \times 10 = 140(N-10)$.

Work ratio = Effort ratio:

$$0.35/0.65 = 70N / 140(N-10)$$

$$0.35 \times 140(N-10) = 0.65 \times 70N$$

$$49(N-10) = 45.5N$$

$$49N - 490 = 45.5N$$

$$3.5N = 490 \rightarrow N = 140.$$

Question 11

Working alone, the times taken by Anu, Tanu and Manu to complete any job are in the ratio 5:8:10. They accept a job which they can finish in 4 days if they all work together for 8 hours per day. However, Anu and Tanu work together for the first 6 days, working 6 hours 40 minutes per day. Then, the number of hours that Manu will take to complete the remaining job working alone is

TITA (Type in the answer)

Solution:

Times ratio 5:8:10 $\rightarrow$ Rates ratio = $1/5 : 1/8 : 1/10 = 8:5:4$ (multiply by 40).

Let rates be $8k, 5k, 4k$ units/hour.

Combined rate = 17k. Total work = $17k \times (4 \text{ days} \times 8 \text{ hrs}) = 17k \times 32 = 544k$ units.

Anu and Tanu work 6 days at 6 hrs 40 min = $20/3$ hrs/day.

Total hours = $6 \times 20/3 = 40$ hours.

Combined rate (Anu+Tanu) = $8k + 5k = 13k$.

Work done = $13k \times 40 = 520k$ units.

Remaining = $544k - 520k = 24k$ units.

Manu's rate = 4k units/hour.

Time for Manu = $24k/4k = 6$ hours.

Question 12

Anil can paint a house in 60 days while Bimal can paint it in 84 days. Anil starts painting and after 10 days, Bimal and Charu join him. Together they complete the painting in 14 more days. If they are paid a total of Rs 21,000 for the job, then the share of Charu, in INR, proportionate to the work done by him, is:

9000) 9000

9200) 9200

9100) 9100

9150) 9150

Solution:

Total work = LCM(60, 84) = 420 units.

Anil's rate = $420/60 = 7$ units/day. Bimal's rate = $420/84 = 5$ units/day.

Anil works alone for 10 days: work done = $10 \times 7 = 70$ units.

Remaining = $420 - 70 = 350$ units.

All three work 14 days. Combined rate = $350/14 = 25$ units/day.

Charu's rate = $25 - 7 - 5 = 13$ units/day.

Work done by each:

Anil: $(10+14) \text{ days} \times 7 = 168$ units.

Bimal: $14 \times 5 = 70$ units.

Charu: $14 \times 13 = 182$ units. Total = 420. Correct.

Charu's share = $(182/420) \times 21000 = 182 \times 50 = \text{Rs } 9100$.

Question 13

Two pipes A and B are attached to an empty water tank. Pipe A fills the tank while pipe B drains it. If pipe A is opened at 2 pm and pipe B is opened at 3 pm, the tank becomes full at 10 pm. Instead, if pipe A is opened at 2 pm and pipe B is opened at 4 pm, the tank becomes full at 6 pm. If pipe B is not opened at all, then the time, in minutes, taken to fill the tank is

TITA (Type in the answer)

Solution:

Let fill rate of A = a units/hr, drain rate of B = b units/hr. Volume = V .

Scenario 1: A works 8 hrs (2-10pm), B works 7 hrs (3-10pm).

$8a - 7b = V \dots(1)$

Scenario 2: A works 4 hrs (2-6pm), B works 2 hrs (4-6pm).

$$4a - 2b = V \dots(2)$$

$$\text{Equate: } 8a - 7b = 4a - 2b \rightarrow 4a = 5b \rightarrow b = (4/5)a.$$

$$\text{Substitute into (2): } V = 4a - 2(4/5)a = 4a - 8a/5 = (20a - 8a)/5 = 12a/5.$$

$$\text{Time for A alone} = V/a = (12a/5)/a = 12/5 \text{ hours} = 12/5 \times 60 = 144 \text{ minutes.}$$

Question 14

One day, Rahul started a work at 9 AM and Gautam joined him two hours later. They then worked together and completed the work at 5 PM the same day. If both had started at 9 AM and worked together, the work would have been completed 30 minutes earlier. Working alone, the time Rahul would have taken, in hours, to complete the work is

11.5) 11.5

10) 10

12.5) 12.5

12) 12

Solution:

Let Rahul's rate = R and Gautam's rate = G (work/hour). Total work = 1.

Case 1: Rahul works 8 hrs (9am-5pm), Gautam works 6 hrs (11am-5pm).

$$8R + 6G = 1 \dots(1)$$

Case 2: Both start at 9am, finish at 4:30pm (30 min earlier = 7.5 hrs).

$$7.5R + 7.5G = 1 \dots(2)$$

$$\text{From (1) - (2): } 0.5R - 1.5G = 0 \rightarrow R = 3G.$$

$$\text{Substitute into (2): } 7.5(3G) + 7.5G = 1 \rightarrow 30G = 1 \rightarrow G = 1/30.$$

$$R = 3/30 = 1/10.$$

$$\text{Time for Rahul alone} = 1/R = 10 \text{ hours.}$$

Question 15

Anil can paint a house in 12 days while Barun can paint it in 16 days. Anil, Barun, and Chandu undertake to paint the house for Rs 24000 and the three of them together complete the painting in 6 days. If Chandu is paid in proportion to the work done by him, then the amount in INR received by him is

TITA (Type in the answer)

Solution:

$$\text{Total work} = \text{LCM}(12, 16) = 48 \text{ units.}$$

$$\text{Anil's rate} = 48/12 = 4 \text{ units/day. Barun's rate} = 48/16 = 3 \text{ units/day.}$$

$$\text{Together they complete in 6 days. Combined rate} = 48/6 = 8 \text{ units/day.}$$

$$\text{Chandu's rate} = 8 - 4 - 3 = 1 \text{ unit/day.}$$

$$\text{Work done in 6 days: Anil} = 24, \text{ Barun} = 18, \text{ Chandu} = 6. \text{ Total} = 48. \text{ Correct.}$$

$$\text{Ratio} = 4:3:1 \text{ (ratio of daily rates, same number of days).}$$

$$\text{Chandu's share} = (1/8) \times 24000 = \text{Rs } 3000.$$

Question 16

The minor angle between the hour hand and the minute hand of a clock was observed at 8:48 am. What is the minimum duration, in minutes, after 8:48 am when this angle increases by 50%?

'36/11) 36/11

2) 2

4) 4

'24/11) 24/11

Solution:

Angle formula: $\theta = |30H - (11/2)M|$ where H = hour, M = minutes.

At 8:48: $\theta = |30(8) - (11/2)(48)| = |240 - 264| = 24$ degrees.

Target angle = $24 \times 1.5 = 36$ degrees.

At 8:48 the minute hand is ahead of the hour hand (minute hand at 288 deg, hour hand at 264 deg).

Relative speed of minute hand w.r.t. hour hand = $6 - 0.5 = 5.5$ degrees/min.

To increase the angle (minute hand moving further ahead):

Additional separation needed = $36 - 24 = 12$ degrees.

Time = $12/5.5 = 12/(11/2) = 24/11$ minutes.

Question 17

The amount of job that Amal, Sunil, and Kamal can individually do in a day are in harmonic progression. Kamal takes twice as much time as Amal to do the same amount of job. If Amal and Sunil work for 4 days and 9 days, respectively, Kamal needs to work for 16 days to finish the remaining job. Then the number of days Sunil will take to finish the job working alone is

TITA (Type in the answer)

Solution:

Let T_A = time for Amal, T_S = time for Sunil, T_K = time for Kamal.

$T_K = 2 \times T_A$ (Kamal takes twice as long as Amal).

Rates in HP means times in AP: $2 \times T_S = T_A + T_K = T_A + 2T_A = 3T_A$.

So $T_S = 1.5 \times T_A$.

Let $T_A = t$. Then $T_S = 1.5t$ and $T_K = 2t$.

Rates: $a = 1/t$, $s = 1/(1.5t) = 2/(3t)$, $k = 1/(2t)$.

Work done: $4a + 9s + 16k = 1$.

$4/t + 9 \times 2/(3t) + 16/(2t) = 1$.

$4/t + 6/t + 8/t = 1$.

$18/t = 1 \rightarrow t = 18$.

$T_S = 1.5 \times 18 = 27$ days.

Question 18

A contractor agreed to construct a 6 km road in 200 days. He employed 140 persons for the work. After 60 days, he realized that only 1.5 km road has been completed. How many additional people would he need to employ in order to finish the work exactly on time?

TITA (Type in the answer)

Solution:

Work proportional to Persons x Days.

Phase 1: 140 persons, 60 days, completed 1.5 km.

Phase 2: P persons, 140 days (remaining), must complete 4.5 km.

Using $(P_1 \times D_1)/W_1 = (P_2 \times D_2)/W_2$:

$$(140 \times 60)/1.5 = (P \times 140)/4.5$$

$$5600 = P \times 140/4.5 \dots \text{re-arrange:}$$

$$(140 \times 60)/1.5 = (P \times 140)/4.5$$

$$60/1.5 = P/4.5$$

$$40 = P/4.5$$

$$P = 40 \times 4.5 = 180 \text{ persons.}$$

$$\text{Additional people needed} = 180 - 140 = 40 \text{ people.}$$

Question 19

Renu would take 15 days working 4 hours per day to complete a certain task whereas Seema would take 8 days working 5 hours per day to complete the same task. They decide to work together to complete this task. Seema agrees to work for double the number of hours per day as Renu, while Renu agrees to work for double the number of days as Seema. If Renu works 2 hours per day, then the number of days Seema will work, is

TITA (Type in the answer)

Solution:

Total work: Renu = $15 \times 4 = 60$ person-hours. Seema = $8 \times 5 = 40$ person-hours. Both equal the same job.

Work = 60 Renu-hours = 40 Seema-hours.

Renu's rate = $1/60$ job/hour. Seema's rate = $1/40$ job/hour.

Let Seema works D_S days. Renu works $2 \times D_S$ days (double days).

Renu works 2 hrs/day. Seema works $2 \times 2 = 4$ hrs/day (double hours).

$$\text{Work by Renu} = (2 \times D_S) \times 2 \times (1/60) = 4D_S/60 = D_S/15.$$

$$\text{Work by Seema} = D_S \times 4 \times (1/40) = 4D_S/40 = D_S/10.$$

$$\text{Total work} = D_S/15 + D_S/10 = 1.$$

$$D_S(1/15 + 1/10) = 1$$

$$D_S(2/30 + 3/30) = 1$$

$$D_S(5/30) = 1$$

$$D_S = 6 \text{ days.}$$

Question 20

John takes twice as much time as Jack to finish a job. Jack and Jim together take one-thirds of the time to finish the job than John takes working alone. Moreover, in order to finish the job, John takes three days more than that taken by three of them working together. In how many days will Jim finish the job working alone?

TITA (Type in the answer)

Solution:

Let Jack's time = J days. John's time = 2J days. Jim's time = M days.

Jack and Jim together take $\frac{1}{3}$ of John's time:

$$\frac{1}{1/J + 1/M} = \frac{2J}{3} \rightarrow \frac{JM}{J+M} = \frac{2J}{3} \rightarrow \frac{M}{J+M} = \frac{2}{3} \rightarrow 3M = 2(J+M) \rightarrow M = 2J.$$

$$\text{All three together: } T_{\text{all}} = \frac{1}{1/J + 1/(2J) + 1/M} = \frac{1}{1/J + 1/(2J) + 1/(2J)} = \frac{1}{((2+1+1)/(2J))} = \frac{2J}{4} = \frac{J}{2}.$$

$$\text{John takes 3 more days than all three together: } 2J = \frac{J}{2} + 3 \rightarrow \frac{3J}{2} = 3 \rightarrow J = 2 \text{ days.}$$

$$\text{Jim's time } M = 2J = 2 \times 2 = 4 \text{ days.}$$

Question 21

A tank has an inlet pipe and an outlet pipe. If the outlet pipe is closed then the inlet pipe fills the empty tank in 8 hours. If the outlet pipe is open then the inlet pipe fills the empty tank in 10 hours. If only the outlet pipe is open then in how many hours the full tank becomes half-full?

A) 20

B) 10

✓ C) 40 ← Correct Answer

D) 15

Solution:

Let tank capacity = 40 units (LCM of 8 and 10).

Inlet rate (outlet closed) = $40/8 = 5$ units/hour.

Net fill rate (both open) = $40/10 = 4$ units/hour.

Outlet drain rate = $5 - 4 = 1$ unit/hour.

To drain half the tank = 20 units.

Time = $20/1 = 40$ hours.
